

# Project Plan: Seatac AI - A Airfield Data Assistant

**Course:** Experience Expo, Fall 2026

**Team Name:** SeaTac AI (Team 3)

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**Instructor:** Prof. David Fannin (Faculty Advisor)

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## 1. Problem Statement

### Current State

SeaTac records detailed data on every aircraft movement through its Surface Area Management System (SAMS): landings, taxi routes, gate arrivals, pushbacks and takeoffs and many more data fields. The data exists perfectly, but leadership can't reach it directly. A question like "Which airline is running late the most?" goes to IT or business intelligence staff, who find the right fields, write the SQL and build a report.

Two earlier cohorts built a prototype chatbot to close this gap. A user types a question, a large language model (LLM) turns it into SQL, the query runs against a one-day sample of SAMS data, and the chatbot replies with text, a table and a chart. Our review of the inherited code shows the prototype can't yet be trusted to give leadership correct answers:

- **Hallucination risk.** The written summary is based on only the first 3 rows of a result. When SQL generation fails, a fallback query returns 10 unrelated flights, which the chatbot then summarizes as if they answered the question.
- **Accuracy.** The model is told about only 4 of the 18 event types and 6 of the 15 flight columns, so it guesses names. It isn't told which of several similar timestamps (scheduled, ATC estimate, actual) to use, or what airport terms like "in-block" and "wheels-up" mean. Any number in a question can become an hour filter, so "top 10 aircraft types" silently returns only 10 a.m. departures. A blanket rule that keeps only times between 1 and 120 minutes also drops early departures, which skews delay figures.
- **Data traps.** `flight_id` is the only unique identifier for a flight, but events aren't linked to it, so they're matched by call sign instead. Call signs repeat (660 for 725 flight legs), so these joins double-count some flights, and the prompt's own example undercounts them. We will add a `flight_id` to every event so each one links to exactly one flight. About 3,200 events that took place at the other airport, judging by their timing and order, are tagged with SEA runways or gates, which inflates runway and gate counts. The sample also has no true scheduled arrival time, so questions about late arrivals need an agreed definition before they can be answered.
- **User experience.** The chatbot never asks a follow-up question. It silently picks a meaning for vague words like "late" or "busiest," forgets context between messages, and sometimes answers real questions with small talk. Error and missing-data messages are unclear, and raw database errors are shown to users as-is, and chart types are guessed from the first row of results.
- **No way to measure.** The README reports 91% accuracy (changed from 78% in a later commit), but the repository has no test set or evaluation script to check either figure.
- **Limited coverage.** Of the Port's 10 priority questions, only 2 have a prebuilt query, prompt guidance and complete data behind them (taxi-in time by aircraft type and taxi-out time by hour).

*These findings come from code review and direct checks of the sample data. We will measure their effect on real answers in a baseline test of the inherited prototype before making changes.*

### Quantifiable Impact

- A BI report for an ad-hoc leadership question takes from about 3 days to 2 weeks, as reported by both earlier cohorts from their work with the Port.
- SeaTac has 89 gates on a site with no room to expand (according to kickoff meeting with Samer Tirhi). An arriving aircraft holding for an occupied gate ties up taxiway space, so leadership needs fast, reliable answers about waiting, delays and gate use.

## 2. SMART Goal Statement

**The Outcome:** By December 2026, Port leadership will be able to ask any of the Port's 10 priority questions in plain English and get a response they can trust: either a correct answer, or a clear explanation of what data or definition is missing. Every answer will state the definition it used.

**Goal:** By December 2026, fix the issues documented in Current State and verify the chatbot on the Port's 10 priority questions. Our initial targets are:

- at least 6 of the 10 questions answered correctly
- the remaining questions clearly explain what data or definition is missing, every time they're asked
- all 8 documented code and data issues fixed and verified

### SMART Criteria

- **Specific:** Fix the documented issues in the existing chatbot and make its answers to the Port's 10 priority questions reliable, without rebuilding it.
- **Measurable:** Each of the 10 questions is asked in two different ways and run three times, then checked against our hand-calculated answer.
- **What counts as correct:** an answer counts only if (1) the calculation matches our hand-checked answer, (2) the written explanation is clear and states the definition used, and (3) any table or chart shows the same numbers as the text.
- **Achievable:** It focuses on 10 questions and the known issues, not new features. It uses the current architecture and the one-day sample the Port confirmed is enough for testing.
- **Relevant:** Accuracy and user experience are the Port's two priorities, and the Port recommended fixing the known issues first.
- **Time-bound:** Baseline by Oct 19, documented issues fixed by Nov 9, final results in December.

## 3. Project Scope

### In-Scope

1. **Test set for the 10 priority questions:** a hand-checked expected answer for each question; a script that asks each one in two different ways and runs it three times; and before/after results.
2. **Data grounding for the LLM:** the full schema, the official SAMS data dictionary, a glossary of airport terms and acronyms, rules for which timestamp to use for each metric, and correct join and location rules.
3. **Reliability fixes:** SELECT-only SQL validation, summaries based on the full result, removal of the misleading fallback answer, fixes for the hour-filter, 1–120 minute filter and small-talk bugs, a `flight_id` link for every event, and a check that the text, table and chart show the same numbers.
4. **Clarification and follow-up (Phase 2):** once the 10 priority questions are reliable, the chatbot asks a short follow-up with 2–4 clickable options plus a free-text "Other" box when a question is ambiguous, and keeps context across turns

5. **Honest limits:** clear replies when the data can't answer a question (e.g., passenger counts, complaints, gate assignments), with suggestions for what it can answer. When missing data affects an answer, the chatbot says what's missing and how many records were left out.
6. **Charts and interface:** chart type chosen from the shape of the data, labelled axes with units, the definition used shown next to each answer, an option to view the SQL, real example questions on the welcome screen, and a cleaner layout.
7. **Data quality and limits report:** profiling of the sample (duplicates, missing values, location tagging, runway events) and a plain-language list of what the data can and can't answer.
8. **Basic security:** read-only database user, removal of exposed credentials, an API key on the backend and restricted CORS.
9. **Handoff documentation:** developer guide, setup guide, glossary and final report for the next cohort.

**Data source:** the provided anonymized sample covering one operating day (August 8, 2025): 725 flight legs, 7,995 flight events and 16 aircraft types. It contains no light aircraft and no true scheduled arrival times, so weight-class and arrival-delay questions have limits.

### Out-of-Scope

- Connecting to live SAMS data or production databases.
- Deploying to Port infrastructure or running in production.
- Migrating to Microsoft Github Copilot (planned for a future cohort).
- Fine-tuning or training model weights.
- Forecasting models. Peak-hour questions will be answered from historical counts in the sample only.
- Automated operational recommendations.
- Single sign-on or full user management.
- The Electron desktop app from the Fall 2025 cohort.
- Adding questions beyond the Port's 10 priority questions until those 10 are verified.

### Technical Choice

The Port asked us to keep the current architecture and improve its accuracy rather than rebuild or migrate, so we are keeping the inherited stack:

| Layer         | Choice                                                                                            | Why                                                                                                                                                                                                    |
|---------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend      | React 18, TypeScript, Vite, Chart.js                                                              | Inherited and working. The team can extend it without a rewrite.                                                                                                                                       |
| Backend       | Python 3.12, FastAPI                                                                              | Inherited. Pydantic models validate every request and response.                                                                                                                                        |
| Database      | MySQL 8.4 with the provided Alplane dump                                                          | Matches the sample data. The app connects through a read-only user.                                                                                                                                    |
| LLM           | Google Gemini 2.5 Flash (development); Anthropic Claude once the Port's API agreement is in place | Gemini support has been added in our local copy and doesn't depend on the pending agreement; it will be committed and tested in Sprint 1. One backend setting switches providers without code changes. |
| Testing       | pytest benchmark runner                                                                           | Scores the chatbot's answers against expected results after every change.                                                                                                                              |
| Collaboration | GitHub (repository, pull requests, Projects board)                                                | Code review, task tracking and the required /docs and /src structure in one place.                                                                                                                     |

The demo runs locally on team laptops; hosting is not required this term. If either model falls short, we switch providers through the same setting (see Risk Register).

## 4. Assumptions and Dependencies

## Assumptions

- The one-day sample would be enough to build and test the prototype (confirmed by the Industry partners at kickoff).
- The Port's 10 priority questions remain the core question set.
- Sending the anonymized sample data to an external LLM API during development is acceptable (to be confirmed with Samer in writing).
- The inherited architecture stays in place for this term.
- Each member can commit about 3–4 hours a week outside class, more around the midterm and final.

## Dependencies

- **From the Port:** which field holds the real schedule for departure delays, and whether a scheduled in-block time exists for arrivals; whether events at the other airport are tagged with SEA locations by design; definitions for terms such as “late” and delay thresholds; confirmation of how runway events are recorded; a mapping of taxiway segment codes to taxiway names; an explanation of duplicate call signs; status of the LLM API agreement; any additional data we justify; a weekly meeting time.
- **From our faculty advisor:** feedback on the plan, benchmark design and midterm, availability and limits of the Gemini and Claude APIs.
- **From the previous cohort (Team 6):** where the 91% accuracy figure came from.

## 5. Stakeholders

### External Stakeholders

| Stakeholder                                     | Role                                                                  | Power  | Interest | How we engage                                                                                     |
|-------------------------------------------------|-----------------------------------------------------------------------|--------|----------|---------------------------------------------------------------------------------------------------|
| Samer Tirhi                                     | Industry partner lead. Defines requirements and accepts deliverables. | High   | High     | Manage closely: weekly Monday meeting, agenda sent the Friday before, demo at each sprint review. |
| Samad Aidane                                    | Senior IT Project Manager. Scheduling, API agreement and data access. | High   | High     | Manage closely: copied on every update; first contact for access and approval issues.             |
| Prof. David Fannin                              | Faculty advisor.                                                      | High   | High     | Manage closely: weekly advisor check-in; review of plan, benchmark and midterm.                   |
| Course instructors (S. Kovala, R. Ma, O. Veliz) | Grading and cross-instructor reviews.                                 | High   | Medium   | Keep satisfied: on-time deliverables; prepared for reviews.                                       |
| Port leadership                                 | End users who ask operational questions.                              | Medium | High     | Keep informed through Samer. The benchmark and demo are built around their questions.             |
| Operations, BI and IT staff                     | Currently answer leadership's data requests; possible test users.     | Low    | Medium   | Keep informed. Invite one or two to the user test through Samer.                                  |
| Port IT/ICT, security and procurement           | Approve API agreements and future live-data access.                   | High   | Low      | Keep satisfied through Samad. No direct contact this term.                                        |
| Previous cohort (Team 6)                        | Built the inherited code.                                             | Low    | Low      | Monitor: one round of knowledge-transfer questions.                                               |

**Internal Stakeholders:** the seven members of Team 3 (see Section 8).

## 6. Business Case (Purpose and Benefits)

### Purpose: Why is this project needed?

SeaTac can't add land or many more gates, so it has to get more out of the space and systems it already has. SAMS already captures the data needed for most operational questions, but only people who know SQL and the data model can use it, so leadership waits on IT and BI for answers. A chatbot that leadership can trust lets them ask directly and get an answer in seconds. Trust is the key word: an answer that looks right but isn't is worse than no answer, which is why this cohort focuses on accuracy before new features.

### Benefits: What is the impact and value?

- **Faster decisions:** supported questions answered in under 30 seconds instead of days or weeks.
- **Less ad-hoc work for BI and IT staff**, who can spend that time on deeper analysis.
- **Consistent answers:** every response states the definition it used, so two people asking the same question see the same logic.
- **Groundwork for later phases:** the benchmark, glossary and validation layer work with any model, so they carry over to live data or a Copilot migration.

## 7. Key Milestones & PM Approach

**Methodology: Hybrid.** Course deadlines are fixed, so we treat them as phase gates (Waterfall). Inside each phase we work in two-week Agile sprints, because requirements will shift as the Port confirms definitions and the baseline shows where the chatbot fails. Tasks live on a GitHub Projects board, and each sprint ends with a review at our Monday meeting with the Port.

### Timeline

| Date          | Milestone                                                                                                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sep 24        | Project launch; first meeting with Samad Aidane.                                                                                                                                           |
| Oct 5         | Kickoff and live operations demo with Samer Tirhi; priorities confirmed.                                                                                                                   |
| <b>Oct 8</b>  | <b>Project Plan due.</b>                                                                                                                                                                   |
| Oct 12        | Sprint 1 ends: every member running the app locally; repository restructured (/docs, /src, AI_USE.md); Gemini provider switch committed and tested; test set for the 10 questions drafted. |
| Oct 15        | Cross-Instructor Team Project Review #1.                                                                                                                                                   |
| Oct 19        | Baseline run of the 10 questions on the inherited prototype; Port definitions confirmed.                                                                                                   |
| Oct 26        | Sprint 2 ends: data grounding, SQL validation and bug fixes merged.                                                                                                                        |
| <b>Oct 29</b> | <b>Midterm Team Presentation: baseline results and first improvements.</b>                                                                                                                 |
| Nov 9         | Sprint 3 ends: all 8 documented issues fixed and verified; clear replies for missing data.                                                                                                 |
| Nov 19        | Cross-Instructor Team Project Review #2.                                                                                                                                                   |
| Nov 23        | Sprint 4 ends: 10 priority questions verified; chart and interface updates; user test with 3–5 non-technical users. Phase 2 starts if time allows.                                         |
| Nov 30        | Code freeze; final benchmark run.                                                                                                                                                          |
| <b>Dec 3</b>  | <b>Client Feedback Report and Executive Summary due.</b>                                                                                                                                   |
| <b>Dec 10</b> | <b>Final Presentation and deliverable handoff to the Port.</b>                                                                                                                             |
| Dec 17        | Expo Week: open expo, poster, project and course reflection.                                                                                                                               |

## 8. Team Members and Roles

| Member                   | Role                                               | Responsibilities                                                                                                                                                                                             |
|--------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Yani Hu                  | Project Manager , Data Quality Development support | Helps with sprint coordination, meeting notes, and project tracking; supports and evaluation; contributes to multi-turn conversation and clarification testing; assists with frontend/backend implementation |
| Vatsalya Rohitbhai Dabhi | Technical Lead and Client Liaison                  | Owns the architecture and LLM integration; reviews and merges pull requests. Single point of contact for Samer and Samad: sends agendas and the weekly status update.                                        |
| Luwei (Helen) Yang       | Backend and Frontend Engineer                      | SQL validation, conversation context, clarification logic and security fixes.                                                                                                                                |
| Tianqing (Jason) Mao     | Backend and Frontend Engineer                      | Interface updates, clickable clarification options, charts and tables.                                                                                                                                       |
| Hanyu (Carl) Xu          | Quality Assurance and Evaluation Lead              | Benchmark design, test runner, baseline and final scoring, failure analysis.                                                                                                                                 |
| Shamali Pachpande        | Data Quality and Domain Lead                       | Data profiling, data dictionary review, airport glossary, timestamp and join rules, data limits report.                                                                                                      |
| Jinghan (Lena) Liang     | UX and User Testing Lead                           | Interaction flows, wording of clarifications and error messages, user test with non-technical users, demo scenarios.                                                                                         |

Expected benchmark answers are calculated independently by the QA Lead and the Data Quality Lead, and any differences are reconciled before scoring.

### Communication Plan

- **Weekly partner meeting:** Mondays with Samer and Samad on Microsoft Teams (1 hour).
- **Weekly status update:** emailed every Friday by the Client Liaison to Samer, Samad and Prof. Fannin: what we finished, what's next, blockers and decisions we need. It doubles as the agenda for Monday.
- **Faculty advisor:** weekly check-in with Prof. Fannin.
- **Team:** Thursday class work session plus one online sync each week; day-to-day discussion in the team chat; every task has an owner on the GitHub board.

## 9. Success Criteria

- **Metric 1:** Reduce the time for leadership to get an answer to a supported question from 3 days–2 weeks (a BI report) to under 30 seconds, resulting in faster operational decisions.
- **Metric 2:** Increase correct answers on the 15 standard benchmark questions from the baseline measured on the inherited prototype (recorded by Oct 19) to at least 12 of 15.
- **Metric 3:** Increase the priority questions we can't fully answer that clearly explain what data or definition is missing, from the measured baseline to all of them, on every run.
- **Metric 4:** Increase the documented code and data issues that are fixed and verified from 0 of 8 to 8 of 8:
  1. the 3-row summary
  2. the silent fallback
  3. the missing schema in the prompt
  4. the hour filter
  5. the 1–120 minute filter
  6. small-talk misclassification
  7. call-sign joins
  8. location tagging
- **Metric 5:** Increase the schema the model is told about from 4 of 18 event types and 6 of 15 flight columns to all 18 and all 15.
- **Metric 6:** Increase the answers to the 10 priority questions that state when missing data affects them (what's missing and how many records were excluded) from the measured baseline to every affected answer.

## 10. Project Risk Register

Severity = Likelihood (1–5) × Impact (1–5).

| Risk Description                                                                                         | Likelihood | Impact | Severity | Contingency Plan (Plan B)                                                                                                                         | Trigger                                                               |
|----------------------------------------------------------------------------------------------------------|------------|--------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| The LLM API agreement is still in progress, and free-tier limits slow testing.                           | 4          | 5      | 20       | Use Gemini's free tier, cache responses and run the full benchmark only at checkpoints; ask our advisor about course credits.                     | Still pending on Oct 26, or the daily quota is hit twice in a week.   |
| Fixes to the inherited code break other features.                                                        | 4          | 4      | 16       | Small reviewed pull requests; rerun the benchmark after each merge; keep the baseline on its own branch.                                          | Score drops after a merge.                                            |
| Our expected answers are wrong.                                                                          | 3          | 5      | 15       | Two members calculate each answer independently.                                                                                                  | The two calculations disagree.                                        |
| Accuracy target not met by December.                                                                     | 3          | 5      | 15       | Focus on the Port's top questions; cut lower-priority interface work; report results honestly.                                                    | Fewer than 5 of the 10 priority questions verified on the Nov 23 run. |
| Key definitions (schedule field, "late," runway events) take longer to confirm than our schedule allows. | 3          | 4      | 12       | Proceed with clearly labelled working definitions and update them once confirmed.                                                                 | Not confirmed by the Oct 19 baseline.                                 |
| Data-handling rules for external AI services turn out stricter than expected.                            | 2          | 5      | 10       | Pause external calls and switch to an approved provider or a locally hosted model.                                                                | Guidance from the Port to stop.                                       |
| A few questions need more than one day of data (e.g., trends, peak-hour patterns).                       | 3          | 3      | 9        | Answer from the sample and state the limit. If a question can't be tested, share specific examples with the Port showing why more data is needed. | A benchmark question can't be validated with one day.                 |
| New requests beyond this term's scope (e.g., live data, Copilot).                                        | 3          | 3      | 9        | Log each request and agree on priorities together at the Monday meeting.                                                                          | A request outside Section 3.                                          |
| A team member is unavailable.                                                                            | 3          | 3      | 9        | A backup owner for each workstream.                                                                                                               | Misses two syncs in a row.                                            |
| The LLM model is changed or retired.                                                                     | 2          | 3      | 6        | Switch models through configuration.                                                                                                              | Deprecation notice or API error.                                      |

**Critical risks (20–25):** The API risk is critical. We are tracking it with Samad and our faculty advisor, and will review it at every Monday meeting until it's resolved.

## 11. Document Repository

Repository URL: <https://github.com/Vatsalya2003/SeaTac-AI>

### Documentation Structure

- **/docs:** Project Plan, meeting minutes, risk register, weekly status updates, project summary, tech stack, database schema guide, developer guide, setup guide, airport glossary, data quality report and benchmark results.
- **/src:** commented source code, split into `/src/frontend` (React), `/src/backend` (FastAPI), `/src/database` (schema dump and import scripts) and `/src/tests` (benchmark questions and runner).

- **AI\_USE.md:** a log of every AI tool used, with the date, tool, task, what it produced and how a team member verified it before use (checking claims against the code and data, reviewing code in pull requests, running the benchmark).
- **README.md:** setup steps and links to the documents above.